Vansh Garg

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Education

Indian Institute of Information Technology Ranchi CGPA: 8.17/10.0
B.Tech in Electronics and Communication Engineering

Work Experience

Electronic Engineering Intern **Jun 2026 – Present**
Recognizant Robotech Engineers Pvt. Ltd.

- Developed networking and wireless communication applications using Python and C++ for defense-oriented engineering assignments.
- Built engineering tools including a Virtual Network Routing Simulator and a Link Budget Calculator for communication system analysis.
- Applied TCP socket programming, routing algorithms, and RF communication concepts to simulate packet forwarding and wireless communication systems.

Personal Projects

HFSS Microstrip Antenna Design Automation using PyAEDT May 2026
Tools: Ansys HFSS, PyAEDT, Python, NumPy, Matplotlib GitHub

- Engineered a fully automated antenna design pipeline using PyAEDT, eliminating manual HFSS GUI workflows.
- Implemented parametric geometry generation with automated simulation setup and frequency sweep.
- Automated extraction of S11, VSWR, Smith Chart, radiation patterns, and bandwidth analysis.
- Designed and optimized a 2.4 GHz microstrip antenna achieving S11 < -10 dB and VSWR < 1.5.

Virtual Network Routing Simulator Jul 2026
Tools: Python, TCP Socket Programming, Dijkstra's Algorithm GitHub

- Developed a virtual network routing simulator enabling communication between multiple interconnected nodes using TCP socket programming.
- Implemented Dijkstra's shortest path algorithm for dynamic route computation and efficient packet forwarding.
- Built a modular routing architecture supporting end-to-end communication, packet forwarding, and shortest-path validation.
- Evaluated routing performance by simulating multi-node network topologies and analyzing packet delivery efficiency.

ESP32-Based Bluetooth Radar System Jan 2025
Tools: ESP32, Arduino IDE, C++, HC-SR04, Servo, Bluetooth Serial, TM1637 GitHub

- Developed a 180° scanning radar system on ESP32 for real-time obstacle detection and ranging.
- Engineered servo-sensor synchronization to map obstacle distance over a polar coordinate grid.
- Integrated TM1637 display, buzzer, and Bluetooth Serial for real-time monitoring and wireless control.
- Optimized ultrasonic timing and servo movement to reduce echo interference and improve measurement accuracy.

Technical Skills

Programming Languages	Python, C++, Embedded C, HTML, CSS
Networking	TCP/IP, Socket Programming, UART, I2C, SPI
Microcontrollers	ESP32, ESP32-S3, STM32, Arduino Uno, Arduino Nano, Arduino Mega, NodeMCU (ESP8266)
RF & Simulation	Ansys HFSS, PyAEDT, LTspice
PCB & Hardware	KiCad, Arduino IDE, JLCPCB, Tinkercad
Frameworks	FastAPI, Django, OpenCV, ROS2 (Fundamentals), NumPy, Matplotlib
Operating Systems	Ubuntu, Arch Linux, Kali Linux, Windows
Concepts	Embedded Systems, IoT, Computer Networks, Antenna Design, RF Fundamentals, PCB Design, Data Structures & Algorithms, OOP

Achievements

- LeetCode: Max Rating 1450+
- Secured 4th Position in Robo Soccer War – Autonomous Robotics Competition.

Positions of Responsibility

Coordinator, IoT & Robotics Wing House of Geeks, IIIT Ranchi

- Organized robotics workshops and technical events for undergraduate students.
- Mentored students in Embedded Systems, Arduino, ESP32 and Robotics.